# Manage S3 audit configurations

ONTAP 9.14.1 REST API reference

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# Manage S3 audit configurations

## Protocols audit svm.uuid object-store endpoint overview

### Overview

S3 events auditing is a security measure that enables you to track and log certain S3 events on storage virtual machines (SVMs). You can track potential security problems and provides evidence of any security breaches.

### Examples

---

#### Creating an S3 audit entry with log rotation size and log retention count

To create an S3 audit entry with log rotation size and log retention count, use the following API. Note the *return_records=true* query parameter is used to obtain the newly created entry in the response.

---

```
# The API:
POST /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X POST "https://<mgmt-ip>/api/protocols/audit/ec650e97-156e-11e9-
abcb-005056bbd0bf/object-store?return_records=true" -H "accept:
application/json" -H "Content-Type: application/json" -d "{ \"enabled\":
true, \"events\": { \"data\": false, \"management\": false}, \"log\": {
\"format\": \"json\", \"retention\": { \"count\": 10 }, \"rotation\": {
\"size\": 2048000 }}, \"log_path\": \"/\"}"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "ec650e97-156e-11e9-abcb-005056bbd0bf",
      "name": "vs1"
    },
    "enabled": true,
    "events": {
      "data": false,
      "management": false
    },
    "log": {
      "format": "json",
      "rotation": {
        "size": 2048000
      },
      "retention": {
        "count": 10,
        "duration": "0s"
      }
    },
    "log_path": "/"
  }
],
"num_records": 1
}
```

#### Creating an S3 audit entry with log rotation schedule and log retention duration

To create an S3 audit entry with log rotation schedule and log retention duration, use the following API. Note that the *return_records=true* query parameter is used to obtain the newly created entry in the response.

```
# The API:
POST /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X POST "https://<mgmt-ip>/api/protocols/audit/a8d64674-13fc-11e9-
87b1-005056a7ae7e/object-store?return_records=true" -H "accept:
application/json" -H "Content-Type: application/json" -d "{ \"enabled\":
false, \"events\": { \"data\": true, \"management\": true }, \"log\": {
\"format\": \"json\", \"retention\": { \"duration\": \"P4DT12H30M5S\" },
\"rotation\": { \"schedule\": { \"days\": [1, 5, 10, 15], \"hours\": [0,
1, 6, 12, 18, 23], \"minutes\": [10, 15, 30, 45, 59], \"months\": [0],
\"weekdays\": [0, 2, 5] } } }, \"log_path\": \"/\"}"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "a8d64674-13fc-11e9-87b1-005056a7ae7e",
      "name": "vs3"
    },
    "enabled": true,
    "events": {
      "data": true,
      "management": true
    },
    "log": {
      "format": "json",
      "rotation": {
        "schedule": {
          "minutes": [
            10,
            15,
            30,
            45,
            59
          ],
          "hours": [
            0,
            1,
            6,
            12,
            18,
            23
          ],
```

```
          "weekdays": [
            0,
            2,
            5
          ],
          "days": [
            1,
            5,
            10,
            15
          ],
          "months": [
            0
          ]
        }
      },
      "retention": {
        "count": 0,
        "duration": "P4DT12H30M5S"
      }
    },
    "log_path": "/"
  }
],
"num_records": 1
}
```

#### Retrieving an S3 audit configuration for all SVMs in the cluster

```
# The API:
GET /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/audit/*/object-
store?fields=*&return_records=true&return_timeout=15" -H "accept:
application/json"

# The response:
{
"records": [
  {
    "svm": {
```

```
      "uuid": "ec650e97-156e-11e9-abcb-005056bbd0bf",
      "name": "vs1"
    },
    "enabled": true,
    "events": {
      "data": false,
      "management": false
    },
    "log": {
      "format": "json",
      "rotation": {
        "size": 2048000
      },
      "retention": {
        "count": 10,
        "duration": "0s"
      }
    },
    "log_path": "/"
  },
  {
    "svm": {
      "uuid": "a8d64674-13fc-11e9-87b1-005056a7ae7e",
      "name": "vs3"
    },
    "enabled": true,
    "events": {
      "data": true,
      "management": true
    },
    "log": {
      "format": "json",
      "rotation": {
        "schedule": {
          "minutes": [
            10,
            15,
            30,
            45,
            59
          ],
          "hours": [
            0,
            1,
            6,
            12,
```

```
            18,
            23
          ],
          "weekdays": [
            0,
            2,
            5
          ],
          "days": [
            1,
            5,
            10,
            15
          ],
          "months": [
            0
          ]
        }
      },
      "retention": {
        "count": 0,
        "duration": "P4DT12H30M5S"
      }
    },
    "log_path": "/"
  }
],
"num_records": 2
}
```


---

**Retrieving specific entries with event list as data and management event for an SVM**

The configuration returned is identified by the events in the list of S3 audit configurations of an SVM.

---

```
# The API:
GET /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/audit/*/object-
store?events.data=true&events.management=true&return_records=true&return_t
imeout=15" -H "accept: application/json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "ec650e97-156e-11e9-abcb-005056bbd0bf",
      "name": "vs1"
    },
    "events": {
      "data": true,
      "management":true
    }
  },
  {
    "svm": {
      "uuid": "a8d64674-13fc-11e9-87b1-005056a7ae7e",
      "name": "vs3"
    },
    "events": {
      "data": true,
      "management": true
    }
  }
],
"num_records": 2
}
```

#### Retrieving a specific S3 audit configuration of an SVM

The configuration returned is identified by the UUID of its SVM.

```
# The API:
GET /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X GET "https://<mgmt-ip>/api/protocols/audit/ec650e97-156e-11e9-
abcb-005056bbd0bf/object-store/" -H "accept: application/json"

# The response:
{
"svm": {
  "uuid": "ec650e97-156e-11e9-abcb-005056bbd0bf",
  "name": "vs1"
},
"enabled": true,
"events": {
  "data": false,
  "management": false
},
"log": {
  "format": "json",
  "rotation": {
    "size": 2048000
  },
  "retention": {
    "count": 10,
    "duration": "0s"
  }
},
"log_path": "/"
}
```

#### Updating a specific S3 audit configuration of an SVM

The configuration is identified by the UUID of its SVM and the provided information is updated.

```
# The API:
PATCH /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X PATCH "https://<mgmt-ip>/api/protocols/audit/ec650e97-156e-11e9-
abcb-005056bbd0bf/object-store/" -H "accept: application/json" -H
"Content-Type: application/json" -d "{ \"enabled\": false}"
```

---

#### Deleting a specific S3 audit configuration of an SVM

The entry to be deleted is identified by the UUID of its SVM.

---

```
# The API:
DELETE /api/protocols/audit/{svm.uuid}/object-store/

# The call:
curl -X DELETE "https://<mgmt-ip>/api/protocols/audit/ec650e97-156e-11e9-
abcb-005056bbd0bf/object-store" -H "accept: application/json"
```

# Delete an S3 audit configuration

DELETE `/protocols/audit/{svm.uuid}/object-store`

**Introduced In:** 9.10

Deletes an S3 audit configuration.

## Related ONTAP commands

- `vserver object-store-server audit disable`
- `vserver object-store-server audit delete`

## Learn more

- DOC /protocols/audit/{svm.uuid}/object-store

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| force | boolean | query | False | Indicates whether a force deletion of the audit configuration is enabled. |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| svm.uuid | string | path | True | UUID of the SVM to which this object belongs. |

## Response

```
Status: 200, Ok
```

## Response

```
Status: 202, Accepted
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 140902420 | Failed to delete audit configuration for the SVM. |
| 140902421 | Failed to delete audit configuration for the SVM because audit is enabled for the SVM. |
| 140902422 | Failed to delete audit configuration for the SVM because final conolidation is in progress. Wait a few mintues, and try the operation again. |

Also see the table of common errors in the Response body overview section of this documentation.

| Name | Type | Description |
|---|---|---|
| error | returned_error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

returned_error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Retrieve an S3 audit configurations

GET `/protocols/audit/{svm.uuid}/object-store`

**Introduced In:** 9.10

Retrieves S3 audit configurations.

## Related ONTAP commands

- `vserver object-store-server audit show`

## Learn more

- DOC /protocols/audit/{svm.uuid}/object-store

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| events.management | boolean | query | False | Filter by events.management |
| events.data | boolean | query | False | Filter by events.data |

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| log_path | string | query | False | Filter by log_path |
| log.rotation.size | integer | query | False | Filter by log.rotation.size |
| log.rotation.schedule.hours | integer | query | False | Filter by log.rotation.schedule.hours<br>• Max value: 23<br>• Min value: 0 |
| log.rotation.schedule.days | integer | query | False | Filter by log.rotation.schedule.days<br>• Max value: 31<br>• Min value: 1 |
| log.rotation.schedule.weekdays | integer | query | False | Filter by log.rotation.schedule.weekdays<br>• Max value: 6<br>• Min value: 0 |
| log.rotation.schedule.minutes | integer | query | False | Filter by log.rotation.schedule.minutes<br>• Max value: 59<br>• Min value: 0 |
| log.rotation.schedule.months | integer | query | False | Filter by log.rotation.schedule.months<br>• Max value: 12<br>• Min value: 1 |
| log.format | string | query | False | Filter by log.format |
| log.retention.count | integer | query | False | Filter by log.retention.count |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| log.retention.duration | string | query | False | Filter by log.retention.duration |
| enabled | boolean | query | False | Filter by enabled |
| svm.name | string | query | False | Filter by svm.name |
| svm.uuid | string | path | True | UUID of the SVM to which this object belongs. |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned.<br>• Default value: 1 |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Max value: 120<br>• Min value: 0<br>• Default value: 1 |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| enabled | boolean | Specifies whether or not auditing is enabled on the SVM. |
| events | events | |
| log | s3_log | |
| log_path | string | The audit log destination path where consolidated audit logs are stored. |
| svm | svm | SVM, applies only to SVM-scoped objects. |

#### Example response


```
{
  "log": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "format": "json",
    "retention": {
      "duration": "P4DT12H30M5S"
    },
    "rotation": {
      "schedule": {
        "days": {
        },
        "hours": {
        },
        "minutes": {
        },
        "months": {
        },
        "weekdays": {
        }
      }
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
| --- | --- | --- |
| error | returned_error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

events

| Name | Type | Description |
|---|---|---|
| data | boolean | Data events |
| management | boolean | Management events |

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

retention

| Name | Type | Description |
|---|---|---|
| count | integer | Determines how many audit log files to retain before rotating the oldest log file out. This is mutually exclusive with "duration". |
| duration | string | Specifies an ISO-8601 format date and time to retain the audit log file. The audit log files are deleted once they reach the specified date/time. This is mutually exclusive with "count". |

audit_schedule

Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.

| Name | Type | Description |
|---|---|---|
| days | array[integer] | Specifies the day of the month schedule to rotate audit log. Leave empty for all. |

| Name | Type | Description |
| --- | --- | --- |
| hours | array[integer] | Specifies the hourly schedule to rotate audit log. Leave empty for all. |
| minutes | array[integer] | Specifies the minutes schedule to rotate the audit log. |
| months | array[integer] | Specifies the months schedule to rotate audit log. Leave empty for all. |
| weekdays | array[integer] | Specifies the weekdays schedule to rotate audit log. Leave empty for all. |

rotation

Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

| Name | Type | Description |
| --- | --- | --- |
| now | boolean | Manually rotates the audit logs. Optional in PATCH only. Not available in POST. |
| schedule | audit_schedule | Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values. |
| size | integer | Rotates logs based on log size in bytes. |

s3_log

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| format | string | Format in which the logs are generated by the consolidation process. Possible values are:<br>• json - ONTAP-specific Json log format.<br>• Default value: 1<br>• enum: ["json"]<br>• Introduced in: 9.10<br>• x-nullable: true |
| retention | retention | |
| rotation | rotation | Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file. |

svm

SVM, applies only to SVM-scoped objects.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. This field cannot be specified in a PATCH method. |
| uuid | string | The unique identifier of the SVM. This field cannot be specified in a PATCH method. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

returned_error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Update an S3 audit configuration for an SVM

PATCH `/protocols/audit/{svm.uuid}/object-store`

**Introduced In:** 9.10

Updates an S3 audit configuration for an SVM.

## Important notes

- `events` - Not specifying either data or management is equivalent to setting it to false.

## Related ONTAP commands

- `vserver object-store-server audit modify`

## Learn more

- DOC /protocols/audit/{svm.uuid}/object-store

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| svm.uuid | string | path | True | UUID of the SVM to which this object belongs. |

## Request Body

| Name | Type | Description |
|---|---|---|
| enabled | boolean | Specifies whether or not auditing is enabled on the SVM. |
| events | events | |
| log | s3_log | |

| Name | Type | Description |
| --- | --- | --- |
| log_path | string | The audit log destination path where consolidated audit logs are stored. |
| svm | svm | SVM, applies only to SVM-scoped objects. |

#### Example request

```
{
  "log": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "format": "json",
    "retention": {
      "duration": "P4DT12H30M5S"
    },
    "rotation": {
      "schedule": {
        "days": {
        },
        "hours": {
        },
        "minutes": {
        },
        "months": {
        },
        "weekdays": {
        }
      }
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

## Response

```
Status: 200, Ok
```

## Response

```
Status: 202, Accepted
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 140902401 | Failed to create an audit configuration for the SVM. |
| 140902402 | Audit configuration is already present. |
| 140902402 | Audit configuration is already enabled. |
| 140902403 | Failed to create staging volume. |
| 140902415 | Failed to modify an audit configuration because no audit configuration exists for SVM. |
| 140902416 | Failed to modify audit configuration for SVM. |
| 140902422 | Final consolidation is in progress, audit delete failed. |
| 140902423 | Failed to delete the audit configuration for the SVM. |
| 140902425 | Audit configuration is not available for disabling. |
| 140902430 | Audit configuration is not available for enabling. |
| 140902431 | Audit enable failed, audit configuration already enabled for the SVM. |
| 140902432 | Final consolidation is in progress, audit enable failed. |
| 140902445 | Audit disable failed, audit configuration does not exist for the SVM. |
| 140902446 | Audit configuration is already disabled. |
| 140902446 | Audit disable failed, audit configuration does not exist for the SVM. |
| 140902456 | The specified log_path does not exist. |
| 140902457 | The log_path must be a directory. |
| 140902458 | The log_path must be a canonical path in the SVM's namespace. |
| 140902459 | The log_path cannot be empty. |
| 140902460 | Rotate size must be greater than or equal to 1024 KB. |
| 140902461 | The destination path must not contain a symbolic link. |

| Error Code | Description |
|---|---|
| 140902470 | The log_path exceeds a maximum supported length of characters. |
| 140902471 | The log_path contains an unsupported read-only (DP/LS) volume. |
| 140902472 | The log_path is not a valid destination for the SVM. |
| 140902474 | The log_path contains an unsupported Snaplock volume. |
| 140902478 | The log_path validation failed. |
| 140902478 | The log_path cannot be accessed for validation. |
| 140902479 | No other fields can be specified when "enabled" is specified. |
| 140902490 | Audit configuration is absent for rotate. |
| 140902491 | Failed to rotate audit log. |
| 140902492 | Cannot rotate audit log, auditing is not enabled for this SVM. |

Also see the table of common errors in the Response body overview section of this documentation. ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 9699340 | SVM UUID lookup failed |
| 9699407 | Additional fields are provided |

| Name | Type | Description |
|---|---|---|
| error | returned_error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

events

| Name | Type | Description |
|---|---|---|
| data | boolean | Data events |
| management | boolean | Management events |

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

retention

| Name | Type | Description |
|---|---|---|
| count | integer | Determines how many audit log files to retain before rotating the oldest log file out. This is mutually exclusive with "duration". |
| duration | string | Specifies an ISO-8601 format date and time to retain the audit log file. The audit log files are deleted once they reach the specified date/time. This is mutually exclusive with "count". |

audit_schedule

Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.

| Name | Type | Description |
|---|---|---|
| days | array[integer] | Specifies the day of the month schedule to rotate audit log. Leave empty for all. |

| Name | Type | Description |
| --- | --- | --- |
| hours | array[integer] | Specifies the hourly schedule to rotate audit log. Leave empty for all. |
| minutes | array[integer] | Specifies the minutes schedule to rotate the audit log. |
| months | array[integer] | Specifies the months schedule to rotate audit log. Leave empty for all. |
| weekdays | array[integer] | Specifies the weekdays schedule to rotate audit log. Leave empty for all. |

rotation

Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

| Name | Type | Description |
| --- | --- | --- |
| now | boolean | Manually rotates the audit logs. Optional in PATCH only. Not available in POST. |
| schedule | audit_schedule | Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values. |
| size | integer | Rotates logs based on log size in bytes. |

s3_log

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| format | string | Format in which the logs are generated by the consolidation process. Possible values are:<br>• json - ONTAP-specific Json log format.<br>• Default value: 1<br>• enum: ["json"]<br>• Introduced in: 9.10<br>• x-nullable: true |
| retention | retention | |
| rotation | rotation | Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file. |

svm

SVM, applies only to SVM-scoped objects.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. This field cannot be specified in a PATCH method. |
| uuid | string | The unique identifier of the SVM. This field cannot be specified in a PATCH method. |

s3_audit

Auditing for NAS events is a security measure that enables you to track and log certain S3 events on SVMs.

| Name | Type | Description |
|---|---|---|
| enabled | boolean | Specifies whether or not auditing is enabled on the SVM. |
| events | events | |
| log | s3_log | |
| log_path | string | The audit log destination path where consolidated audit logs are stored. |
| svm | svm | SVM, applies only to SVM-scoped objects. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

returned_error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Create an S3 audit configuration

POST `/protocols/audit/{svm.uuid}/object-store`

**Introduced In:** 9.10

Creates an S3 audit configuration.

## Required properties

- `log_path` - Path in the owning SVM namespace that is used to store audit logs.

## Default property values

If not specified in POST, the following default property values are assigned:

- `enabled` - *true*
- `events.data` - *true*
- `events.management` - *false*
- `log.format` - *json*
- `log.retention.count` - *0*
- `log.retention.duration` - *PT0S*
- `log.rotation.size` - *100MB*
- `log.rotation.now` - *false*

## Related ONTAP commands

- `vserver object-store-server audit create`
- `vserver object-store-server audit enable`

## Learn more

- DOC /protocols/audit/{svm.uuid}/object-store

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| svm.uuid | string | path | True | UUID of the SVM to which this object belongs. |

## Request Body

| Name | Type | Description |
|---|---|---|
| enabled | boolean | Specifies whether or not auditing is enabled on the SVM. |
| events | events | |
| log | s3_log | |

| Name | Type | Description |
| --- | --- | --- |
| log_path | string | The audit log destination path where consolidated audit logs are stored. |
| svm | svm | SVM, applies only to SVM-scoped objects. |

**Example request**

```
{
  "log": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "format": "json",
    "retention": {
      "duration": "P4DT12H30M5S"
    },
    "rotation": {
      "schedule": {
        "days": {
        },
        "hours": {
        },
        "minutes": {
        },
        "months": {
        },
        "weekdays": {
        }
      }
    }
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

## Response

```
Status: 202, Accepted
```

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| num_records | integer | Number of records |
| records | array[s3_audit] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "log": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "format": "json",
      "retention": {
        "duration": "P4DT12H30M5S"
      },
      "rotation": {
        "schedule": {
          "days": {
          },
          "hours": {
          },
          "minutes": {
          },
          "months": {
          },
          "weekdays": {
          }
        }
      }
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
```

```
      }
    }
  }
```

**Headers**

| Name | Description | Type |
|---|---|---|
| Location | Useful for tracking the resource location | string |

## Response

```
Status: 201, Created
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 140902401 | Failed to create an audit configuration for the SVM. |
| 140902402 | Audit configuration is already present. |
| 140902402 | Audit configuration is already enabled. |
| 140902403 | Failed to create staging volume. |
| 140902415 | Failed to modify an audit configuration because no audit configuration exists for the SVM. |
| 140902416 | Failed to modify audit configuration for SVM. |
| 140902422 | Final consolidation is in progress, audit delete failed. |
| 140902423 | Failed to delete the audit configuration for the SVM. |
| 140902425 | Audit configuration is not available for disabling. |
| 140902430 | Audit configuration is not available for enabling. |
| 140902431 | Audit enable failed, audit configuration already enabled for the SVM. |
| 140902432 | Final consolidation is in progress, audit enable failed. |
| 140902445 | Audit disable failed, audit configuration does not exist for the SVM. |

| Error Code | Description |
| --- | --- |
| 140902446 | Audit disable failed, audit configuration does not exist for the SVM. |
| 140902447 | Audit disable failed. |
| 140902456 | The specified log_path does not exist. |
| 140902457 | The log_path must be a directory. |
| 140902458 | The log_path must be a canonical path in the SVM's namespace. |
| 140902459 | The log_path cannot be empty. |
| 140902460 | Rotate size must be greater than or equal to 1024 KB. |
| 140902461 | The destination path must not contain a symbolic link. |
| 140902470 | The log_path exceeds a maximum supported length of characters. |
| 140902471 | The log_path contains an unsupported read-only (DP/LS) volume. |
| 140902472 | The log_path is not a valid destination for the SVM. |
| 140902474 | The log_path contains an unsupported Snaplock volume. |
| 140902478 | The log_path validation failed. |
| 140902478 | The log_path cannot be accessed for validation. |
| 140902490 | Audit configuration is absent for rotate. |
| 140902491 | Failed to rotate audit log. |
| 140902492 | Cannot rotate audit log, auditing is not enabled for this SVM. |

Also see the table of common errors in the Response body overview section of this documentation. ONTAP Error Response Codes

| Error Code | Description |
| --- | --- |
| 9699340 | SVM UUID lookup failed |
| 9699407 | Additional fields are provided |

| Name | Type | Description |
| --- | --- | --- |
| error | returned_error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

events

| Name | Type | Description |
|---|---|---|
| data | boolean | Data events |
| management | boolean | Management events |

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

retention

| Name | Type | Description |
|---|---|---|
| count | integer | Determines how many audit log files to retain before rotating the oldest log file out. This is mutually exclusive with "duration". |
| duration | string | Specifies an ISO-8601 format date and time to retain the audit log file. The audit log files are deleted once they reach the specified date/time. This is mutually exclusive with "count". |

audit_schedule

Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values.

| Name | Type | Description |
|---|---|---|
| days | array[integer] | Specifies the day of the month schedule to rotate audit log. Leave empty for all. |

| Name | Type | Description |
| --- | --- | --- |
| hours | array[integer] | Specifies the hourly schedule to rotate audit log. Leave empty for all. |
| minutes | array[integer] | Specifies the minutes schedule to rotate the audit log. |
| months | array[integer] | Specifies the months schedule to rotate audit log. Leave empty for all. |
| weekdays | array[integer] | Specifies the weekdays schedule to rotate audit log. Leave empty for all. |

rotation

Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file.

| Name | Type | Description |
| --- | --- | --- |
| now | boolean | Manually rotates the audit logs. Optional in PATCH only. Not available in POST. |
| schedule | audit_schedule | Rotates the audit logs based on a schedule by using the time-based rotation parameters in any combination. The rotation schedule is calculated by using all the time-related values. |
| size | integer | Rotates logs based on log size in bytes. |

s3_log

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| format | string | Format in which the logs are generated by the consolidation process. Possible values are:<br>• json - ONTAP-specific Json log format.<br>• Default value: 1<br>• enum: ["json"]<br>• Introduced in: 9.10<br>• x-nullable: true |
| retention | retention | |
| rotation | rotation | Audit event log files are rotated when they reach a configured threshold log size or are on a configured schedule. When an event log file is rotated, the scheduled consolidation task first renames the active converted file to a time-stamped archive file, and then creates a new active converted event log file. |

svm

SVM, applies only to SVM-scoped objects.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. This field cannot be specified in a PATCH method. |
| uuid | string | The unique identifier of the SVM. This field cannot be specified in a PATCH method. |

s3_audit

Auditing for NAS events is a security measure that enables you to track and log certain S3 events on SVMs.

| Name | Type | Description |
| --- | --- | --- |
| enabled | boolean | Specifies whether or not auditing is enabled on the SVM. |
| events | events | |
| log | s3_log | |
| log_path | string | The audit log destination path where consolidated audit logs are stored. |
| svm | svm | SVM, applies only to SVM-scoped objects. |

_links

| Name | Type | Description |
| --- | --- | --- |
| next | href | |
| self | href | |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

returned_error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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